

Problem 21. There is a pile of 1000 matches. Two players each take turns and can take 1 to 5 matches. It is also allowed at most 10 times during the whole game to take 6 matches. For example, 7 exceptional moves can be done by the first player and 3 moves by the second and then no more exceptional moves are allowed. Whoever takes the last match wins. Determine which player has a winning strategy.

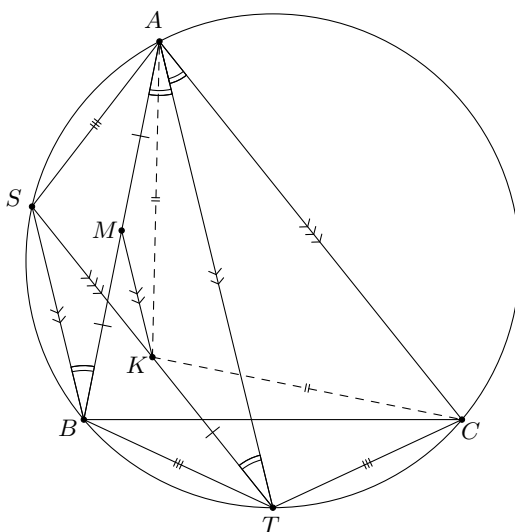
Solution. The second player wins.

Let r be the number of the remaining exceptional moves in the current position (at the beginning of the game $r = 10$ and r decreases during the game). The winning strategy of the second player is the following. After his move the number of matches in the pile must have the form $6n + r$, where $n > r$, or $7n$, where $n \leq r$ (observe that $6n + r = 7n$ for $n = r$). At the beginning of the game the initial number of matches $1000 = 6 \cdot 165 + 10$ agrees with this strategy. What happens during two consecutive moves? Consider the case $n > r$ first. If the first player takes $k = 1, 2, \dots, 5$ matches (and hence r is not changing during his move) then the second player takes $6 - k$ matches. So players take 6 matches together and the pile contains now $6(n - 1) + r$ matches. If the first player takes 6 matches, then r decreases by 1. The second player takes 1 match. After his turn the pile contains $6(n - 1) + (r - 1)$ matches as he wish. Now consider the case $n \leq r$. In this situation we have much enough exceptional moves, and we may assume that now each move the players can take up to 6 matches. Then if the first player takes k matches, the second player takes $7 - k$ matches. $\square$

Problem 22. To every vertex of a regular pentagon a real number is assigned. We may perform the following operation repeatedly: we choose two adjacent vertices of the pentagon and replace each of the two numbers assigned to these vertices by their arithmetic mean. Is it always possible to obtain the position in which all five numbers are zeroes, given that in the initial position the sum of all five numbers is equal to zero?

Solution. We will show that starting from the numbers $-\frac{1}{5}, -\frac{1}{5}, -\frac{1}{5}, -\frac{1}{5}, \frac{4}{5}$, we cannot get five zeroes. By adding $\frac{1}{5}$ to all vertices we see that our task is equivalent to showing that beginning from numbers $0, 0, 0, 0, 1$ and performing the same operations we can never get five numbers $\frac{1}{5}$. This we prove by noticing that in the initial position all the numbers are "binary rational" – that is, of the form $\frac{k}{2^m}$, where k is an integer and m is a non-negative integer – and an arithmetic mean of two binary rationals is also such a number, while the number $\frac{1}{5}$ is not of such form. $\square$

Problem 23. Triangle ABC is given. Let M be the midpoint of the segment AB and T be the midpoint of arc BC not containing A of the circumcircle of $\triangle ABC$. The point K inside $\triangle ABC$ is such that $MATK$ is an isosceles trapezoid with $AT \parallel MK$. Show that $AK = KC$.



Solution. Assume that TK intersects the circumcircle of $\triangle ABC$ at the point S (where $S \neq T$). Then $\angle ABS = \angle ATS = \angle BAT$, so $ASBT$ is a trapezoid. Hence, $MK \parallel AT \parallel SB$ and M is the midpoint of AB . Thus, K is the midpoint of TS . From $\angle TAC = \angle BAT = \angle ATS$ we see that $ACTS$ is an inscribed trapezoid, so it is isosceles. Thus, $AK = KC$, since K is the midpoint of TS . $\square$

Problem 24. For a positive integer k , let $d(k)$ denote the number of divisors of k (e.g. $d(12) = 6$) and let $s(k)$ denote the digit sum of k (e.g. $s(12) = 3$). A positive integer n is said to be *amusing* if there exists a positive integer k such that $d(k) = s(k) = n$. What is the smallest amusing odd integer greater than 1?

Solution. The answer is 9. For every k we have $s(k) \equiv k \pmod{9}$. Calculating remainders modulo 9 we have the following table

m	0	1	2	3	4	5	6	7	8
m^2	0	1	4	0	7	7	0	4	1
m^6	0	1	1	0	1	1	0	1	1

If $d(k) = 3$, then $k = p^2$ with p a prime, but $p^2 \equiv 3 \pmod{9}$ is impossible. This shows that 3 is not an amusing number. If $d(k) = 5$, then $k = p^4$ with p a prime, but $p^4 \equiv 5 \pmod{9}$ is impossible. This shows that 5 is not an amusing number. If $d(k) = 7$, then $k = p^6$ with p a prime, but $p^6 \equiv 7 \pmod{9}$ is impossible. This shows that 7 is not an amusing number. To see that 9 is amusing, note that $d(36) = s(36) = 9$. $\square$